B. And It's Non-Zero

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

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You are given an array consisting of all integers from [l, r] inclusive. For example, if $l = 2$ and $r = 5$, the array would be [2, 3, 4, 5]. What's the minimum number of elements you can delete to make the bitwise AND of the array non-zero?

A *bitwise AND* is a binary operation that takes two equal-length binary representations and performs the *AND* operation on each pair of the corresponding bits.

Input

The first line contains one integer t ($1 \le t \le 10^4$) — the number of test cases. Then t cases follow.

The first line of each test case contains two integers l and r ($1 \le l \le r \le 2 \cdot 10^5$) — the description of the array.

Output

For each test case, output a single integer — the answer to the problem.

Example

Input
5 1 2 2 8 4 5 1 5 100000 200000

Output
1 3 0 2 31072

Note

In the first test case, the array is [1, 2]. Currently, the *bitwise AND* is 0, as 1 & 2 = 0. However, after deleting 1 (or 2), the array becomes [2] (or [1]), and the *bitwise AND* becomes 2 (or 1). This can be proven to be the optimal, so the answer is 1.

In the second test case, the array is [2, 3, 4, 5, 6, 7, 8]. Currently, the *bitwise AND* is 0. However, after deleting 4, 5, and 8, the array becomes [2, 3, 6, 7], and the *bitwise AND* becomes 2. This can be proven to be the optimal, so the answer is 3. Note that there may be other ways to delete 3 elements.